GCAM V8.0

Practical Working & Knowledge Transfer (KT)

DOCUMENT PURPOSE
This document explains how the GCAM V8.0 system works in practical terms, what each part does, how to operate it, and how to hand the system over to a new technician/operator.

Item	Details
Project	GCAM – RealTech Systems
Build reviewed	GCAM-BUILD V8.0
Target platform	Raspberry Pi 5 – 1 GB profile
Main application	Python application under /opt/gcam
Web service	Local dashboard on port 8080
Document type	Practical Working + KT
Theme	Red & White

IMPORTANT: Passwords, MQTT tokens and Cloudflare tunnel tokens are intentionally not printed in this KT document. Keep secrets in the approved secure location.

1. What is GCAM?

GCAM is a Raspberry Pi based smart camera monitoring system. The Pi receives video from an IP camera, performs local processing/AI, provides a web dashboard, stores event recordings, sends selected information to the server using MQTT/SFTP, and can be made reachable from outside the site using a permanent Cloudflare Tunnel.

Function	Simple meaning
Live Camera	View the camera from the local dashboard or public Cloudflare URL.
Person Detection	Detect a person in the configured area and trigger the configured warning/event process.
Vehicle Detection	Detect cars, buses, motorbikes/two-wheelers in the configured vehicle area.
Garbage Detection	Use AI mode or normal/reference-based mode to identify garbage changes/objects.
Event Image	Capture a high-quality image when an event is confirmed.
Event Video	Keep a short pre/post event clip and upload it.
Audio	Play warning/default/received audio through the Pi audio output.
MQTT	Receive commands and publish device/events/status.
SFTP	Upload images/videos/audio to the central file server.
Device Health	Show temperature, network, RAM, battery voltage/current and service-related status.
Remote Access	Cloudflare provides the permanent public web entry point.

ONE-LINE UNDERSTANDING
Camera → Raspberry Pi → AI / Recording / Audio → MQTT + SFTP + Dashboard → Cloudflare remote access.

2. Main System Architecture

The practical data flow is:

Step	What happens
1	IP camera produces MAIN and SUB RTSP streams.
2	Raspberry Pi receives the streams over the local LAN.
3	SUB stream is used for AI processing and live-view work.
4	MAIN stream is used for high-quality event still images and event-video capture.
5	GCAM application runs the detection, event logic, dashboard and storage functions.
6	Confirmed events can be saved locally and uploaded to the SFTP server.
7	MQTT carries device events, commands and responses.
8	Cloudflare Tunnel exposes the Pi web application through the permanent hostname.
9	A Cloudflare watchdog checks the local application, tunnel connector and public URL.

IMPORTANT STREAM RULE
Do not swap the stream roles casually. The build is designed so the SUB stream is used for AI/live viewing, while the MAIN stream is used for higher-quality event image/video uploads.

3. Hardware – Practical Understanding

Hardware	Purpose	Technician check
IP CCTV Camera	Provides video over RTSP.	Camera power, LAN link, IP and RTSP stream.
Raspberry Pi 5	Main GCAM processing computer.	Power, temperature, storage, network and services.
Router / Internet	Connects the site to the internet.	LAN gateway and WAN availability.
INA226 via I2C	Measures battery voltage/current for health display.	I2C wiring/address and reading.
Audio output	Plays warning and other audio.	Correct ALSA output/device and speaker/amplifier.
SD card	OS, application and local playback/event storage.	Free space; avoid filling the card.

4. Software Components

Component	Role
gcam.service	Main GCAM application. Runs app.py and restarts automatically if it exits.
gcam-controller.service	MQTT command controller. Listens for approved remote commands.
cloudflared.service	Permanent Cloudflare Tunnel connector.
gcam-cloudflare-watchdog.timer	Regularly checks Cloudflare connector/public URL and can restart the tunnel.
gcam-app-restart.timer	Scheduled application restart every 3 hours.
gcam-maintenance.timer	Twice-daily maintenance: package cleanup, journal cleanup and sync.
gcam-pi-reboot.timer	Optional twice-daily full Pi reboot; default installer setting is disabled.
/opt/gcam/data/config.json	Runtime configuration: camera, network, MQTT, SFTP, Cloudflare and feature settings.
/opt/gcam/data/state.json	Current device/system/event state used by the application/dashboard.
/opt/gcam/files/recordings	Local storage-based playback recordings.
/opt/gcam/logs	GCAM logs and Cloudflare watchdog log.

5. Camera Stream Working

MAIN stream

- Used for high-quality event still images.
- Used for event-video capture/upload.
- The installer specifically rejects a SUB stream when entered as the MAIN stream.

SUB stream

- Used for AI detection and normal live viewing.
- Public Cloudflare live quality is intentionally conservative for the Pi 5 1 GB profile.
- The build avoids unnecessarily opening another camera session for playback by reusing the local shared live feed where possible.

WHY TWO STREAMS?
AI processing needs a lighter stream to reduce CPU load. Event evidence needs better quality. Splitting the work between SUB and MAIN gives a practical balance on a low-RAM Pi.

6. AI & Detection – Human-Level Explanation

Person detection

- The system uses a MobileNet-SSD person model plus additional filtering logic.
- A person is not treated as an event merely because a single frame contains a person.
- The person must satisfy the configured area/geofence and remain stable for the configured stay time. Default stay time in V8.0 is 5 seconds.
- When the event is confirmed, the system can create event evidence and play the person warning audio, subject to the configured warning-audio rules.
- Person event video is designed around a 6-second pre-event + 8-second post-event window (14 seconds total).

Vehicle detection

- Uses MobileNet-SSD vehicle classes: car, bus, motorbike and bicycle; the UI labels bicycle-type detection as Two-Wheeler.
- Default mode is STATIC. MOVING mode is also available.
- The vehicle area is controlled using a configurable geofence/polygon.
- Default static stay time is 5 seconds. A short detection flicker is tolerated so a parked vehicle does not immediately lose its state.
- The build also supports a second vehicle event while the vehicle remains in the marked area.

Garbage detection

- Two modes are available: AI and NORMAL.
- AI mode uses the supplied custom garbage_yolo.onnx model.
- Normal/reference mode compares the current scene with a stored reference and applies filtering so normal scene changes are less likely to be treated as newly added garbage.
- The garbage process supports manual detection, sample capture and reference-image upload from the dashboard/API.
- Garbage counts are grouped into LOW, MEDIUM and HIGH ranges; the build defines the thresholds as 1+, 3+ and 6+ detected items respectively.

7. What Happens When an Event Occurs?

Stage	Practical action
Detect	AI/normal logic examines frames from the configured area.
Confirm	Time/geofence/confidence/filter rules decide whether it is a real event.
Capture	Event image is prepared from the MAIN stream for quality.
Record	Short event video is assembled using the rolling buffer and post-event frames.

Audio	Configured warning audio may play when the event rule allows it.
Save	Files are stored under the relevant /opt/gcam/files area.
Upload	SFTP sends event media to the configured central folder.
Notify	MQTT can publish the event/status information.
Dashboard	State and latest event information become visible in the web UI.

EVENT EVIDENCE
The build intentionally keeps high-quality event still images separate from the lighter AI/live stream. If the MAIN stream is temporarily stale, the application waits/attempts a direct MAIN capture rather than silently using a low-quality SUB frame.

8. Audio & Live Talk

- GCAM creates a warning.wav during installation using espeak-ng and ffmpeg.
- The warning message in the supplied installer is: “Warning. Person detected. Please leave the monitored area.”
- Audio files can be handled through device-specific Audio and Default_Audio SFTP folders.
- The system supports an audio queue and queue-clear control.
- Live Talk is provided through the web interface/API; audio is processed in short chunks and played on the Pi.
- The build checks real internet connectivity for network-dependent audio behaviour and can manage volume accordingly.

FIELD CHECK
If the dashboard says an audio action happened but there is no sound, check: amplifier/speaker power → Pi audio device → volume → audio process → network/audio source if applicable.

9. Web Dashboard & Useful Pages

Page / API	Purpose
/	Main dashboard.
/live	Live-only view.
/playback	Recorded playback interface.
/live_talk	Live Talk interface.
/camera	Camera configuration proxy.
/garbage_geofencing	Garbage area setup.
/person_geofencing	Person area setup.
/vehicle_geofencing	Vehicle area setup.
/api/state	Overall device/system state.
/api/storage	Storage information.
/api/controls/feature	Enable/disable detection or recording features.
/api/controls/garbage/manual_detect	Manual garbage detection.
/api/controls/garbage/reference_upload	Upload/update garbage reference.
/api/controls/device/restart	Request device restart.

LOCAL ACCESS
The GCAM application listens on port 8080. Local dashboard format: http://<PI-IP>:8080/

10. Remote Access – Cloudflare

- A permanent named Cloudflare Tunnel can map the public hostname to the local GCAM web service at http://localhost:8080.
- The installer stores the tunnel token in /etc/gcam/cloudflared.token with restricted permissions.
- The permanent web base URL is https://<subdomain>.<domain>.
- Useful public paths include /live, /playback and /live_talk.
- A separate temporary Cloudflare camera URL function can be created on demand for camera configuration access.
- The watchdog checks three things: local GCAM health, Cloudflare connector readiness and public hostname health.
- After repeated failures, the watchdog can restart cloudflared, with a cooldown to prevent restart loops.

CLOUDFLARE ERROR 1033
In practical terms, first check that the hostname is routed to the correct Cloudflare tunnel, the token belongs to that tunnel,

cloudflared.service is running, and outbound connectivity (including port 7844) is available.

11. MQTT – Practical Working

MQTT is the remote message channel. The GCAM controller subscribes to the configured command topic and the application publishes responses/events on the configured topics.

Topic type	Purpose
<DeviceID>/command	Remote commands to the Pi.
<DeviceID>/device_response	Response/status from the Pi.
<DeviceID>/camera_event	Camera/person/vehicle related event messages.
<DeviceID>/garbage_level	Garbage detection/level messages.

Important supported commands

Command	What it does
Close The Software	Stops gcam.service.
Open The Software	Starts gcam.service.
Cloudflare Restart	Restarts the permanent Cloudflare connector.
Cloudflare Status	Returns Cloudflare status information.
Camera Permanent URL	Returns the permanent camera configuration URL.
Camera Temp URL	Creates a temporary camera URL when supported.
Person Detection Enable / Disable	Turns person detection on/off.
Vehicle Detection Enable / Disable	Turns vehicle detection on/off.
Garbage Detection Enable / Disable	Turns garbage detection on/off.
Person/Vehicle Video Recording Enable / Disable	Controls event-video recording.
Power Saving / Status	Controls/checks power-saving behaviour.

MQTT SAFETY
The controller validates the configured command token. Retained MQTT controller commands are intentionally ignored/cleared so an old command is not automatically executed again after reconnect.

12. SFTP – Where Event Files Go

Folder purpose	Default pattern
Main device folder	/ftp/files/<DeviceID>
Person images	/ftp/files/<DeviceID>/Person
Person videos	/ftp/files/<DeviceID>/Video/Person_Video
Vehicle images	/ftp/files/<DeviceID>/Vehicle
Vehicle videos	/ftp/files/<DeviceID>/Video/Vehicle_Video
Garbage	/ftp/files/<DeviceID>/Garbage
Device Audio	/ftp/files/<DeviceID>/Audio
Default Audio	/ftp/files/Default_Audio

- The installer requires real Linux-style absolute SFTP paths.
- SFTP upload failures are tracked and retried by the application.
- For a field technician, the most important check is whether the SFTP server is reachable and whether new files appear in the correct DeviceID folder.

13. Local Recording & Playback

- Playback recording is enabled in the V8.0 configuration.
- The recorder uses the shared SUB live feed to avoid creating an unnecessary second camera RTSP session.
- Recording segments are 10 seconds in the configuration.
- The installer protects a minimum SD free-space reserve; the default is 3 GB and the allowed range is 3–5 GB.
- Playback provides play/pause, forward/reverse, speed control, seek and fullscreen controls.

- A clean reinstall deletes the previous /opt/gcam/files/recordings directory before rebuilding the application.

SD CARD RULE
Never allow the SD card to become full. If storage is critically low, playback/event recording can be affected. Check /api/storage and the filesystem before troubleshooting AI problems.

14. Device Health – What to Watch

Health item	What it tells you
Camera connected	Whether the Pi is receiving camera frames.
Last frame time	Whether the camera feed is updating.
SFTP status	Last successful/failing upload state.
MQTT status	Last successful/failing broker communication.
Network status / interface	LAN connectivity and active interface.
Network RX/TX	Traffic activity.
Pi temperature	Thermal condition.
Battery voltage/current	INA226 readings when hardware is present/configured.
RAM usage	Memory pressure on the 1 GB Pi.
Last error	Latest important application error.

15. Normal Startup / Daily Procedure

1. Power ON the camera, router and Raspberry Pi.
2. Wait for the Pi to obtain network connectivity.
3. Allow systemd to start gcam.service and gcam-controller.service automatically.
4. Confirm the camera feed is updating.
5. Open the local dashboard: http://<PI-IP>:8080/
6. Check Camera Connected, temperature, RAM, network, MQTT and SFTP status.
7. Open the Live page and confirm the live picture is stable.
8. If remote access is required, check the permanent Cloudflare URL.
9. If the site uses event monitoring, perform one controlled test for person/vehicle/garbage and confirm the expected image/video/MQTT/audio response.

DO NOT RESTART FIRST
When something fails, check status and logs first. Restarting without collecting the error can hide the original cause.

16. Troubleshooting – Quick KT

Problem	First checks	Typical action
No dashboard	Pi reachable? gcam.service active?	Check service status and app log; restart gcam only after checking the error.
No camera image	Camera power/IP/RTSP; last_frame_at.	Test camera locally and verify RTSP URLs.
AI not detecting	Feature enabled? Correct SUB stream? Geofence?	Check dashboard settings and recent app logs.
Wrong/low event image quality	MAIN RTSP correct?	Confirm MAIN URL is a MAIN stream, not SUB.
No person warning	Person feature/audio enabled? Stable stay met?	Check warning audio status and person settings.
No vehicle event	Vehicle mode/geofence/stay time.	Check static/moving mode and polygon.
Garbage not detected	Garbage mode/reference/model.	Check AI/Normal mode, reference image and manual detect.
No MQTT	Broker/network/credentials/topic.	Check MQTT state and controller log.
No SFTP upload	Server/path/credentials/network.	Check SFTP error state and remote folder.
Public URL fails	cloudflared + DNS/tunnel route.	Check cloudflared status/log and watchdog log.
Pi running hot	CPU load/stream load/fan/heatsink.	Check temperature and reduce unnecessary processing.
SD storage full	/api/storage / filesystem.	Remove/rotate old recordings using approved procedure; keep reserve.

17. Technician Command Sheet

These are the most useful commands for first-line support.

```
# Main application
sudo systemctl status gcam.service --no-pager
sudo systemctl restart gcam.service
journalctl -u gcam.service -n 100 --no-pager

# MQTT controller
sudo systemctl status gcam-controller.service --no-pager
sudo systemctl restart gcam-controller.service
journalctl -u gcam-controller.service -n 100 --no-pager

# Cloudflare
sudo systemctl status cloudflared.service --no-pager
sudo systemctl restart cloudflared.service
sudo journalctl -u cloudflared.service -n 200 --no-pager

# Cloudflare watchdog
sudo systemctl status gcam-cloudflare-watchdog.timer --no-pager
sudo journalctl -u gcam-cloudflare-watchdog.service -n 100 --no-pager
sudo tail -n 100 /opt/gcam/logs/cloudflare_watchdog.log

# All GCAM timers
systemctl list-timers --all | grep gcam

# Useful files
sudo nano /opt/gcam/data/config.json
sudo nano /opt/gcam/data/state.json
```

18. KT – Knowledge Transfer Section

A new technician should be able to explain and perform the following without developer assistance.

KT topic	Person receiving KT should be able to…
System purpose	Explain what GCAM monitors and why the Pi is installed.
Hardware	Identify camera, Pi, router, audio path and battery sensor.
Streams	Explain why SUB is used for AI/live and MAIN for high-quality event evidence.
Dashboard	Open the dashboard and understand the main health indicators.
Detection	Explain person, vehicle and garbage event flow at a basic level.
MQTT	Know where commands enter and where responses/events are published.
SFTP	Find the correct Person/Vehicle/Garbage/Audio folders.
Cloudflare	Check whether the permanent remote URL is working.
Services	Check gcam, controller and cloudflared services.
Logs	Collect the correct journal logs before restarting.
Storage	Check SD free space and understand the protected reserve.
Audio	Check the physical audio path and software status.
Recovery	Restart only the affected service first; reboot the Pi only when necessary.
Escalation	Know when to stop troubleshooting and escalate with logs + screenshots.

18.1 KT Practical Test

10. Show the trainee the camera, Pi, router and audio hardware.
11. Ask the trainee to open the local dashboard.
12. Ask them to identify Camera Connected, temperature, RAM, MQTT and SFTP status.
13. Demonstrate one controlled detection event.
14. Show where the resulting image/video is stored locally and on SFTP.
15. Disconnect/recover a non-critical network path and show how to read the status.
16. Ask the trainee to check gcam.service and read its last 100 log lines.
17. Ask the trainee to check cloudflared.service and the watchdog log.
18. Explain which credentials are secret and where they are securely stored.
19. Record the trainee name/date/sign-off in the project handover record.

19. Preventive Maintenance Checklist

Frequency	Check
Daily / shift	Dashboard opens; live image stable; camera connected; no obvious errors.
Daily / shift	Pi temperature and RAM are reasonable.
Daily / shift	SFTP and MQTT last-success indicators are healthy.
Daily / shift	Remote Cloudflare URL works if remote monitoring is required.
Weekly	Check SD free space and event/playback behaviour.
Weekly	Check camera lens, mounting and field of view.
Weekly	Check Pi fan/heatsink and power connection.
Monthly	Review recurring service/log errors and network reliability.
After changes	Run a controlled person/vehicle/garbage test and verify SFTP/MQTT/audio.

20. Important Build Defaults / Reference

Setting	Build reference
Pi local web port	8080
Detection processing width	640 px
Local live width	960 px
Public Cloudflare live width	640 px
Local live target FPS	10
Public Cloudflare target FPS	3
Person default static stay	5 sec
Vehicle default static stay	5 sec
Garbage default mode	AI
Garbage scheduled interval	3600 sec / 1 hour
Event recording window	6 sec pre + 8 sec post
Playback segment	10 sec
Protected SD reserve	3 GB default; 3–5 GB accepted
App scheduled restart	Every 3 hours
Maintenance	02:45 and 14:45
Full Pi reboot timer	Disabled by default
Cloudflare protocol	Auto by default

CHANGE CONTROL
Do not change AI resolution, thresholds, RTSP roles, timers or network settings casually. V8.0 contains several compatibility/tuning values that are intended to keep the Pi 5 1 GB deployment stable.

21. Escalation Information to Collect

Before escalating a problem to the developer, collect:

- Device ID / camera name.
- Approximate time the problem started.
- Local dashboard screenshot showing health state.
- Whether local live view works.
- Whether the public Cloudflare URL works.
- gcam.service status and last 100 log lines.
- gcam-controller.service status/log if MQTT is involved.
- cloudflared.service status/log if remote access is involved.
- SFTP/MQTT error shown on dashboard.
- Pi temperature, RAM and SD free space.
- A short description of exactly what was expected versus what actually happened.

22. One-Page Quick Reference

NORMAL FLOW
CAMERA → Pi → SUB for AI/LIVE + MAIN for EVENT QUALITY → LOCAL DASHBOARD → MQTT/SFTP → CLOUDFLARE REMOTE ACCESS

Need to check	Command / URL
Local dashboard	http://<PI-IP>:8080/
Live	http://<PI-IP>:8080/live
App service	sudo systemctl status gcam.service --no-pager
App log	journalctl -u gcam.service -n 100 --no-pager
Controller	sudo systemctl status gcam-controller.service --no-pager
Controller log	journalctl -u gcam-controller.service -n 100 --no-pager
Cloudflare	sudo systemctl status cloudflared.service --no-pager
Cloudflare log	sudo journalctl -u cloudflared.service -n 200 --no-pager
Watchdog log	sudo tail -n 100 /opt/gcam/logs/cloudflare_watchdog.log
Timers	systemctl list-timers --all | grep gcam
Config	/opt/gcam/data/config.json
State	/opt/gcam/data/state.json
Recordings	/opt/gcam/files/recordings

Golden Rules

- Check before restarting.
- Keep MAIN and SUB RTSP roles correct.
- Never expose or paste passwords/tokens into general KT documents.
- Keep the SD card from becoming full.
- If remote access fails, check local GCAM first, then Cloudflare.
- If MQTT fails, check network/broker/controller before changing the application.
- After any configuration change, perform one controlled end-to-end test.